

ECE 310 – WIRELESS COMMUNICATION

Milestone 1 (M1) Report

Group 05

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Title: Open-Set / “Unknown Signal” Aware AI Classifier for GNU Radio

Category: GSoC (Google Summer of Code) – AI Spectrum in GNU Radio

1 Problem Statement

1.1 The Gap We Found

While studying existing AI tools that work with GNU Radio, such as **gr-phasma** and **TorchSig**, we noticed that every AI-based signal classifier we found is a *closed-set classifier*. Which means that the model is only trained on a fixed list of signal types which are BPSK, QPSK, AM, FM. If we give this kind of model a signal it has never seen before, or even plain noise or garbage, it will still be forced to output from guess. It never states “I don’t know what this is.”

Which is clearly a weakness in the wider area of modulation classification, where it is called open-set recognition. Many recent papers study this problem in theory, but we could not find a ready, plug-and-play GNU Radio block that actually does this inside a real flowgraph. Every practical GNU Radio tool we found is still closed-set.

1.2 Problem Statement

How can we build a simple GNU Radio block that classifies a radio signal into one of the known modulation types, but also correctly flags the signal as “Unknown” when the model is not confident about its answer, instead of always forcing a guess?

1.3 Motivation

In the real radio spectrum, a receiver will always encounter some signals that were never part of its training data, such as a new device, an unfamiliar modulation scheme, jamming, or plain background noise. A cognitive radio or spectrum monitoring system that is forced to label every such signal as something it already knows is misleading. It can lead to wrong decisions, for example, ignoring an unknown/dangerous interferer because it was wrongly labelled as a harmless known signal. Adding the ability to say “Unknown” makes the classifier safer and more trustworthy for real-world use.

1.4 Significance

- It fixes a real, practical gap in the current open-source GNU Radio AI tooling, not just a theoretical one.
- It is directly useful for cognitive radio, spectrum monitoring, and interference/threat detection use cases, where unseen signals are common.

- It gives a clear, visual, and convincing demo: known signals get correctly classified, and an unseen signal or noise gets flagged in red as “Unknown”, instead of being silently misclassified.

1.5 Assumptions

- We will work mainly in simulation using GNU Radio Companion (GRC), file sources, and signal generators. We do not assume access to physical SDR hardware, although we may be using that later if time allows.
- We are here assuming that RadioML dataset or something similar gives us enough labelled examples of “known” modulation types to train a small baseline classifier.

1.6 Scope

We will training a small CNN classifier on a limited number of known modulation classes from RadioML, adding a rejection rule on top of it to flag low-confidence outputs as “Unknown”, packaging this as a GNU Radio out-of-tree (OOT) block, and showing the result as a colour-coded overlay (green = known, red = unknown) on GNU Radio’s existing waterfall/QT GUI display.

2 Review of Existing Work

Our literature survey is divided into two sections:

- (a) available practical, open-source tools for GNU Radio / RF signal classification
- (b) recent academic papers on open-set recognition in the field of modulation classification which explores the exact problem that we wish to solve.

2.1 Existing Open-Source Tools (GNU Radio Ecosystem)

Tool	What it does	Method	Open-set?
gr-phasma	Modulation Classification Out-of-Tree Module for GNU Radio.	Blocks for detection/classification using signal processing & ML algorithms.	No – closed-set only
TorchSig / Sig53	Open-source RF machine-learning framework that includes a GNU Radio block (specDetect) and a big dataset (Sig53) for classification of narrowband and wideband signals.	CNN / transformer / YOLO models trained on Sig53.	No – fixed 53 class label set

Tool	What it does	Method	Open-set?
RadioML dataset	Open public data set with simulated modulated RF signals, developed by DeepSig/T. O’Shea to train classifiers.	Simulated IQ data with different types of modulation and SNR values.	N/A – dataset, not a classifier
gr-inspector	GNU Radio signal-analysis tool for detection and separation of existing signals within wideband spectrum.	Signal detection based on energy and feature extraction. Detection of unknown signals without modulation level rejection/classification.	Detects unknown signals, but does not classify or reject at the modulation level

From the above table, it can be seen that the practical, ready-to-use GNU Radio tools (gr-phasma, TorchSig) have robust architectures but are closed-set; this means that regardless of the nature of the input signal, they will output one of the trained labels.

2.2 Academic Research on Open-Set Recognition for Modulation Classification

Research-wise, the problem of “open-set” automatic modulation classification (“AMOSR - Automatic Modulation Open-Set Recognition”) is a relatively new trend that keeps developing:

- A pioneering paper by Wu et al. describes the first AMOSR solution, namely the DC-LSTM model which uses both center loss and a statistical distribution (Weibull). Their experiments proved that adding the open-set handling to the classification task helps considerably when unknown modulations appear in comparison to a regular closed-set model. ([arXiv:2002.12037](#))
- The survey paper dedicated to open-set recognition of communication signal modulation claims that AMOSR (identification of known classes and unknown modulation signals) remains under-researched compared to the regular closed-set AMC problem. (Y. Yang et al., IEEE TPAMI 2023)
- In more recent research (2023–2025), the solutions become even heavier, including such complex approaches as the Class Information Guided Reconstruction framework and Class Feature Space Reconstruction (CFSR). These approaches use both reconstruction losses and specially designed feature spaces to both perform the classification and identify unknown signals successfully. ([arXiv:2312.13023](#); IEEE Xplore, 2025)
- Other works consider incorporating deep learning with expertly constructed features in order to spot some signals that would not be picked up by a pure neural network and even consider attacks against open-set classifiers which further proves that this area is developing rapidly and there are many open questions to be answered in it. ([arXiv:2302.03749](#))

2.3 What We Learned from the Review

- The concept of “open-set” recognition for modulation classification is widely explored in scientific papers but has not made its way into the real-life GNU Radio blocks yet.

- Most academic approaches for open-set classification (CFSR, class-information-guided reconstruction, GAN-based approaches) while being fairly accurate, are rather computationally expensive in terms of development and training which complicates their implementation within a short period of time.
- An easier approach based on a rejection rule with a threshold or confidence score calculated using a previously built and understood CNN classifier is a computationally less expensive approach which is also a good benchmark that is used multiple times in these research papers.

3 Establishing our State-of-the-Art (SOTA) Position

According to our research, we see our project in an intersection of two state-of-the-art tracks:

- **State-of-the-art best practical/open-source tools track:** Currently, TorchSig with its Sig53 dataset and GNU Radio block (specDetect) is the best and the most complete ready-to-use open-source AI signal classifier for GNU Radio. The downside of TorchSig is that it works in a closed-set way.
- **State-of-the-art research track:** CFSR and class-information guided reconstruction are among the methods which currently show the best accuracy of open-set modulation recognition in scientific papers. Nevertheless, they are still research codes, there are no blocks available to use directly in a GNU Radio flowgraph.

The state-of-the-art of our project lies in the fact that there is no open-source GNU Radio block which unites both these qualities - the plug-and-play functionality and the ability to classify unknown signals in an appropriate way without guessing their type. We are not going to outperform the accuracy scores of the most sophisticated research works (CFSR, for example).

Instead, we aim at introducing the concept of open set rejection based on confidence-thresholds – which is widely used as a baseline for comparisons in academic research – to an actual GNU Radio-compatible block, and showing through our experiments that it outperforms naively chosen baselines such as the closed set classifier that just guesses, or even just an energy threshold.

As we progress to the next milestones, we will continually verify our status as SOTA: if time permits after creating the initial working prototype, we could consider moving away from the plain softmax-confidence threshold to something a little better, like something along the lines of the center-loss/Weibull-based solution seen in DC-LSTM.

4 Timeline

Our timeline below covers the project from the initial proposal through Milestone 1 (M1) submission, and gives a plan for the milestones after it.

Week	Task	Output
Week 1	Analyze GNU Radio codebase; compile GNU Radio from sources; investigate GNU Radio Modtool; understand GNU Radio blocks & flowgraphs; find issues in the community.	Initial project proposal

Week	Task	Output
Week 2	Refine the problem statement; analyse the existing GNU Radio AI solutions (gr-phasma, TorchSig), review the relevant AMC research papers; determine the SOTA position.	This M1 report (Problem statement, literature review, SOTA position, timeline)
Week 3	Generate a subset of RadioML data; develop a baseline classifier based on a convolutional neural network for a subset of modulation types.	Trained baseline classifier + training script
Week 4	Implement the confidence threshold rejection mechanism as a layer above the baseline classifier; test the system offline for known vs. unknown/modulation classes.	Open-set logic module + offline test results
Week 5	Develop GNU Radio OOT (out of tree) block which will include the classifier and the rejection mechanism; incorporate into GRC flowgraph.	Working GNU Radio block (prototype v1)
Week 6	Develop the colour-coding overlay (green for known, red for unknown) for the GNU Radio Waterfall/QT display; conduct the known-vs-unknown demonstration.	Visual demo flowgraph (.grc) with live colour-coded output
Week 7	Compare our solution to the naive one (classifier with closed set recognition and energy threshold detector); gather the results of testing.	Comparison experiment and results
Week 8	Complete the documentation (README.md, comments, instructions); generate the PDF, video demonstration, Google Forms.	Milestone submission package (PDF, video, code, forms)

Weeks 1–2 above correspond to the work already completed for this M1 submission. Weeks 3 onward are our planned path for the next milestones, and will be refined further based on feedback from our course mentor and, where possible, GNU Radio community mentors.

5 References

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